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DBMS WEEK 5, LAB 4

[TASK 1] – required schema

```
PES2UG24CS337@root [(none)]>use hostel
Database changed
PES2UG24CS337@root [hostel]>CREATE TABLE Hostel (
    ->     Hostel_ID INT PRIMARY KEY,
    ->     Hostel_Name VARCHAR(100) NOT NULL,
    ->     Location VARCHAR(100) NOT NULL,
    ->     Capacity INT NOT NULL DEFAULT 0,
    ->     CONSTRAINT chk_hostel_capacity CHECK (Capacity >= 0)
    -> );
Query OK, 0 rows affected (0.04 sec)

PES2UG24CS337@root [hostel]>CREATE TABLE Staff (
    ->     Staff_ID INT PRIMARY KEY,
    ->     Staff_Name VARCHAR(100) NOT NULL,
    ->     Salary DECIMAL(10,2) NOT NULL,
    ->     Designation VARCHAR(50) NOT NULL DEFAULT 'Support Staff',
    ->     Hostel_ID INT NOT NULL,
    ->     Supervisor_ID INT NULL,
    ->     CONSTRAINT chk_staff_salary CHECK (Salary > 0),
    ->     CONSTRAINT fk_staff_hostel
    ->         FOREIGN KEY (Hostel_ID) REFERENCES Hostel(Hostel_ID),
    ->     CONSTRAINT fk_staff_supervisor
    ->         FOREIGN KEY (Supervisor_ID) REFERENCES Staff(Staff_ID)
    ->         ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
PES2UG24CS337@root [hostel]>INSERT INTO Hostel (Hostel_ID, Hostel_Name, Location, Capacity) VALUES
    -> (1, 'Sunrise Hostel', 'North Campus', 120),
    -> (2, 'Blue Ridge Hostel', 'South Campus', 90),
    -> (3, 'Lake View Hostel', 'East Campus', 60);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

PES2UG24CS337@root [hostel]>INSERT INTO Staff
    -> (Staff_ID, Staff_Name, Designation, Salary, Hostel_ID, Supervisor_ID) VALUES
    -> (1, 'Ramesh Iyer', 'Warden', 45000, 1, NULL),
    -> (2, 'Sunil Kumar', 'Security', 22000, 1, 1),
    -> (3, 'Geeta Rao', 'Cleaner', 18000, 1, 1),
    -> (4, 'Anita Sharma', 'Warden', 46000, 2, NULL),
    -> (5, 'Mohan Das', 'Security', 21000, 2, 4),
    -> (6, 'Priya Menon', 'Maintenance', 25000, 2, 4),
    -> (7, 'Kiran Bose', 'Warden', 44000, 3, NULL),
    -> (8, 'Farhan Sheikh', 'Cleaner', 17500, 3, 7);
Query OK, 8 rows affected (0.01 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

```
PES2UG24CS337@root [hostel]>
PES2UG24CS337@root [hostel]>SELECT * FROM Hostel;
+-----------+-------------------+--------------+----------+
| Hostel_ID | Hostel_Name       | Location     | Capacity |
+-----------+-------------------+--------------+----------+
|         1 | Sunrise Hostel    | North Campus |      120 |
|         2 | Blue Ridge Hostel | South Campus |       90 |
|         3 | Lake View Hostel  | East Campus  |       60 |
+-----------+-------------------+--------------+----------+
3 rows in set (0.00 sec)

PES2UG24CS337@root [hostel]>SELECT * FROM Staff;
+----------+---------------+----------+-------------+-----------+---------------+
| Staff_ID | Staff_Name    | Salary   | Designation | Hostel_ID | Supervisor_ID |
+----------+---------------+----------+-------------+-----------+---------------+
|        1 | Ramesh Iyer   | 45000.00 | Warden      |         1 |          NULL |
|        2 | Sunil Kumar   | 22000.00 | Security    |         1 |             1 |
|        3 | Geeta Rao     | 18000.00 | Cleaner     |         1 |             1 |
|        4 | Anita Sharma  | 46000.00 | Warden      |         2 |          NULL |
|        5 | Mohan Das     | 21000.00 | Security    |         2 |             4 |
|        6 | Priya Menon   | 25000.00 | Maintenance |         2 |             4 |
|        7 | Kiran Bose    | 44000.00 | Warden      |         3 |          NULL |
|        8 | Farhan Sheikh | 17500.00 | Cleaner     |         3 |             7 |
+----------+---------------+----------+-------------+-----------+---------------+
8 rows in set (0.00 sec)

PES2UG24CS337@root [hostel]>SELECT * FROM Room;
+-----------+-------------+-------+
| Hostel_ID | Room_Number | Floor |
+-----------+-------------+-------+
|         1 |         101 |     1 |
|         1 |         102 |     1 |
|         1 |         201 |     2 |
|         2 |         101 |     1 |
|         2 |         102 |     1 |
|         3 |         101 |     1 |
+-----------+-------------+-------+
6 rows in set (0.00 sec)
```

[TASK 2] – part A: DML and set operations

1.
```
                              INSERT INTO Student
    -> (Student_ID, First_Name, Middle_Name, Last_Name, Gender, DOB, Email)
    -> VALUES
    -> (9, 'Kabir', 'S', 'Oberoi', 'Male', '2005-08-19', 'kabir.oberoi@pes.edu');
Query OK, 1 row affected (0.01 sec)
```

2.
```
PES2UG24CS337@root [hostel]>INSERT INTO Student_Phone (Student_ID, Phone_Number) VALUES
    -> (9, '9876500009'),
    -> (9, '9876500019');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

3.
```
PES2UG24CS337@root [hostel]>INSERT INTO Uses (Student_ID, Facility_ID, Usage_Date) VALUES
    -> (9, 1, '2026-08-22'),
    -> (9, 5, '2026-08-23'),
    -> (5, 3, '2026-08-22');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

4.
```
PES2UG24CS337@root [hostel]>UPDATE Staff
    -> SET Salary = Salary * 1.10
    -> WHERE Designation = 'Warden';
Query OK, 3 rows affected (0.03 sec)
Rows matched: 3  Changed: 3  Warnings: 0
```

5.
```
PES2UG24CS337@root [hostel]>UPDATE Allocation
    -> SET Check_out_Date = '2026-08-20'
    -> WHERE Student_ID = 3
    ->   AND Hostel_ID = 1
    ->   AND Room_Number = 102
    ->   AND Check_out_Date IS NULL;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

6.
```
PES2UG24CS337@root [hostel]>UPDATE Bed
    -> SET Status = 'Vacant'
    -> WHERE Hostel_ID = 1
    ->   AND Room_Number = 102
    ->   AND Bed_Number = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

7.
```
PES2UG24CS337@root [hostel]>INSERT INTO Payment
    -> (Payment_ID, Student_ID, Payment_Date, Amount, Payment_Type)
    -> VALUES
    -> (9, 9, '2026-08-22', 45000, 'Online');
Query OK, 1 row affected (0.01 sec)
```

8.
```
PES2UG24CS337@root [hostel]>INSERT INTO Receipt
    -> (Receipt_ID, Payment_ID, Receipt_Date)
    -> VALUES
    -> (8, 9, '2026-08-22');
Query OK, 1 row affected (0.01 sec)
```

9.
```
PES2UG24CS337@root [hostel]>DELETE FROM Payment
    -> WHERE Payment_ID NOT IN (
    ->     SELECT Payment_ID
    ->     FROM Receipt
    -> );
Query OK, 1 row affected (0.01 sec)
```

10.
```
PES2UG24CS337@root [hostel]>SELECT Staff_Name AS Name, 'Staff' AS Role
    -> FROM Staff
    -> WHERE Hostel_ID = 1
    ->
    -> UNION ALL
    ->
    -> SELECT CONCAT(First_Name, ' ', Last_Name) AS Name, 'Student' AS Role
    -> FROM Student
    -> WHERE Student_ID IN (
    ->     SELECT Student_ID
    ->     FROM Allocation
    ->     WHERE Hostel_ID = 1
    ->       AND Check_out_Date IS NULL
    -> );
+---------------+---------+
| Name          | Role    |
+---------------+---------+
| Ramesh Iyer   | Staff   |
| Sunil Kumar   | Staff   |
| Geeta Rao     | Staff   |
| Aditya Verma  | Student |
| Neha Kulkarni | Student |
+---------------+---------+
5 rows in set (0.01 sec)
```

11.
```
PES2UG24CS337@root [hostel]>SELECT Student_ID
    -> FROM Payment
    ->
    -> INTERSECT
    ->
    -> SELECT Student_ID
    -> FROM Uses;
+------------+
| Student_ID |
+------------+
|          1 |
|          2 |
|          3 |
|          4 |
|          5 |
|          6 |
|          7 |
|          9 |
+------------+
8 rows in set (0.00 sec)
```

12.
```
PES2UG24CS337@root [hostel]>SELECT Student_ID
    -> FROM Student
    ->
    -> EXCEPT
    ->
    -> SELECT Student_ID
    -> FROM Payment;
+------------+
| Student_ID |
+------------+
|          8 |
+------------+
1 row in set (0.00 sec)
```

[TASK 2] – part B: Join queries

1.
```
PES2UG24CS337@root [hostel]>SELECT
    ->     s.Student_ID,
    ->     s.First_Name,
    ->     s.Middle_Name,
    ->     s.Last_Name,
    ->     h.Hostel_Name,
    ->     a.Room_Number
    -> FROM Student s
    -> JOIN Allocation a
    ->     ON s.Student_ID = a.Student_ID
    -> JOIN Hostel h
    ->     ON a.Hostel_ID = h.Hostel_ID
    -> WHERE a.Check_out_Date IS NULL;
+------------+------------+-------------+-----------+-------------------+-------------+
| Student_ID | First_Name | Middle_Name | Last_Name | Hostel_Name       | Room_Number |
+------------+------------+-------------+-----------+-------------------+-------------+
|          1 | Aditya     | NULL        | Verma     | Sunrise Hostel    |         101 |
|          2 | Neha       | R           | Kulkarni  | Sunrise Hostel    |         101 |
|          4 | Sanya      | K           | Malhotra  | Blue Ridge Hostel |         101 |
|          5 | Vikram     | NULL        | Nair      | Blue Ridge Hostel |         102 |
|          6 | Ishita     | P           | Bansal    | Blue Ridge Hostel |         102 |
|          7 | Arjun      | NULL        | Reddy     | Lake View Hostel  |         101 |
+------------+------------+-------------+-----------+-------------------+-------------+
6 rows in set (0.00 sec)
```

2.

```
PES2UG24CS337@root [hostel]>SELECT
    ->     s.Staff_ID,
    ->     s.Staff_Name,
    ->     sup.Staff_Name AS Supervisor_Name
    -> FROM Staff s
    -> LEFT JOIN Staff sup
    ->     ON s.Supervisor_ID = sup.Staff_ID;
+----------+---------------+-----------------+
| Staff_ID | Staff_Name    | Supervisor_Name |
+----------+---------------+-----------------+
|        1 | Ramesh Iyer   | NULL            |
|        2 | Sunil Kumar   | Ramesh Iyer     |
|        3 | Geeta Rao     | Ramesh Iyer     |
|        4 | Anita Sharma  | NULL            |
|        5 | Mohan Das     | Anita Sharma    |
|        6 | Priya Menon   | Anita Sharma    |
|        7 | Kiran Bose    | NULL            |
|        8 | Farhan Sheikh | Kiran Bose      |
+----------+---------------+-----------------+
8 rows in set (0.00 sec)
```

3.

```
PES2UG24CS337@root [hostel]>SELECT
    ->     p.Payment_ID,
    ->     CONCAT(s.First_Name, ' ', s.Last_Name) AS Student_Name,
    ->     p.Payment_Date,
    ->     p.Amount,
    ->     p.Payment_Type,
    ->     r.Receipt_ID
    -> FROM Payment p
    -> JOIN Student s
    ->     ON p.Student_ID = s.Student_ID
    -> LEFT JOIN Receipt r
    ->     ON p.Payment_ID = r.Payment_ID;
+------------+----------------+--------------+----------+--------------+------------+
| Payment_ID | Student_Name   | Payment_Date | Amount   | Payment_Type | Receipt_ID |
+------------+----------------+--------------+----------+--------------+------------+
|          1 | Aditya Verma   | 2026-01-10   | 45000.00 | Online       |          1 |
|          2 | Neha Kulkarni  | 2026-01-11   | 45000.00 | Cash         |          2 |
|          3 | Rohan Shetty   | 2026-01-12   | 42000.00 | Card         |          3 |
|          4 | Sanya Malhotra | 2026-01-10   | 48000.00 | Online       |          4 |
|          5 | Vikram Nair    | 2026-01-13   | 48000.00 | Cheque       |          5 |
|          6 | Ishita Bansal  | 2026-01-14   | 48000.00 | Online       |          6 |
|          7 | Arjun Reddy    | 2026-01-15   | 40000.00 | Cash         |          7 |
|          9 | Kabir Oberoi   | 2026-08-22   | 45000.00 | Online       |          8 |
+------------+----------------+--------------+----------+--------------+------------+
8 rows in set (0.00 sec)
```

4.

```
PES2UG24CS337@root [hostel]>SELECT
    ->     s.Student_ID,
    ->     s.First_Name,
    ->     s.Last_Name
    -> FROM Student s
    -> LEFT JOIN Uses u
    ->     ON s.Student_ID = u.Student_ID
    -> WHERE u.Student_ID IS NULL;
+------------+------------+-----------+
| Student_ID | First_Name | Last_Name |
+------------+------------+-----------+
|          8 | Meera      | Pillai    |
+------------+------------+-----------+
1 row in set (0.00 sec)
```

5.

```
PES2UG24CS337@root [hostel]>SELECT
    ->     h.Hostel_ID,
    ->     h.Hostel_Name,
    ->     COUNT(a.Student_ID) AS Current_Residents
    -> FROM Hostel h
    -> LEFT JOIN Allocation a
    ->     ON h.Hostel_ID = a.Hostel_ID
    ->    AND a.Check_out_Date IS NULL
    -> GROUP BY h.Hostel_ID, h.Hostel_Name;
+-----------+------------------+-------------------+
| Hostel_ID | Hostel_Name      | Current_Residents |
+-----------+------------------+-------------------+
|         1 | Sunrise Hostel   |                 2 |
|         2 | Blue Ridge Hostel |                3 |
|         3 | Lake View Hostel  |                1 |
+-----------+------------------+-------------------+
3 rows in set (0.00 sec)
```

6.

```
PES2UG24CS337@root [hostel]>SELECT
    ->     f.Facility_Name,
    ->     CONCAT(s.First_Name, ' ', s.Last_Name) AS Student_Name,
    ->     u.Usage_Date
    -> FROM Uses u
    -> JOIN Facility f
    ->     ON u.Facility_ID = f.Facility_ID
    -> JOIN Student s
    ->     ON u.Student_ID = s.Student_ID
    -> ORDER BY f.Facility_Name, u.Usage_Date;
+---------------+---------------+------------+
| Facility_Name | Student_Name  | Usage_Date |
+---------------+---------------+------------+
| Common Room   | Arjun Reddy   | 2026-01-23 |
| Common Room   | Kabir Oberoi  | 2026-08-23 |
| Gym           | Sanya Malhotra | 2026-01-20 |
| Gym           | Vikram Nair   | 2026-01-21 |
| Gym           | Vikram Nair   | 2026-08-22 |
| Laundry Room  | Aditya Verma  | 2026-01-21 |
| Study Room    | Ishita Bansal | 2026-01-22 |
| Study Room    | Sanya Malhotra | 2026-01-25 |
| WiFi Lounge   | Aditya Verma  | 2026-01-20 |
| WiFi Lounge   | Neha Kulkarni | 2026-01-20 |
| WiFi Lounge   | Rohan Shetty  | 2026-01-22 |
| WiFi Lounge   | Kabir Oberoi  | 2026-08-22 |
+---------------+---------------+------------+
12 rows in set (0.00 sec)
```

7.
```
PES2UG24CS337@root [hostel]>SELECT
    ->     r.Hostel_ID,
    ->     r.Room_Number,
    ->     COUNT(CASE WHEN b.Status = 'Vacant' THEN 1 END) AS Vacant_Beds
    -> FROM Room r
    -> LEFT JOIN Bed b
    ->     ON r.Hostel_ID = b.Hostel_ID
    ->    AND r.Room_Number = b.Room_Number
    -> GROUP BY r.Hostel_ID, r.Room_Number
    -> ORDER BY r.Hostel_ID, r.Room_Number;
+-----------+-------------+-------------+
| Hostel_ID | Room_Number | Vacant_Beds |
+-----------+-------------+-------------+
|         1 |         101 |           0 |
|         1 |         102 |           2 |
|         1 |         201 |           1 |
|         2 |         101 |           1 |
|         2 |         102 |           0 |
|         3 |         101 |           1 |
+-----------+-------------+-------------+
6 rows in set (0.00 sec)
```

8.
```
PES2UG24CS337@root [hostel]>SELECT
    ->     s.Staff_ID,
    ->     s.Staff_Name,
    ->     s.Designation,
    ->     h.Hostel_Name,
    ->     h.Location
    -> FROM Staff s
    -> JOIN Hostel h
    ->     ON s.Hostel_ID = h.Hostel_ID;
+----------+---------------+-------------+-------------------+--------------+
| Staff_ID | Staff_Name    | Designation | Hostel_Name       | Location     |
+----------+---------------+-------------+-------------------+--------------+
|        1 | Ramesh Iyer   | Warden      | Sunrise Hostel    | North Campus |
|        2 | Sunil Kumar   | Security    | Sunrise Hostel    | North Campus |
|        3 | Geeta Rao     | Cleaner     | Sunrise Hostel    | North Campus |
|        4 | Anita Sharma  | Warden      | Blue Ridge Hostel | South Campus |
|        5 | Mohan Das     | Security    | Blue Ridge Hostel | South Campus |
|        6 | Priya Menon   | Maintenance | Blue Ridge Hostel | South Campus |
|        7 | Kiran Bose    | Warden      | Lake View Hostel  | East Campus  |
|        8 | Farhan Sheikh | Cleaner     | Lake View Hostel  | East Campus  |
+----------+---------------+-------------+-------------------+--------------+
8 rows in set (0.00 sec)
```

9.
```
PES2UG24CS337@root [hostel]>SELECT
    ->     h.Hostel_ID,
    ->     h.Hostel_Name,
    ->     COALESCE(SUM(p.Amount), 0) AS Total_Fee_Revenue
    -> FROM Hostel h
    -> LEFT JOIN Allocation a
    ->     ON h.Hostel_ID = a.Hostel_ID
    ->    AND a.Check_out_Date IS NULL
    -> LEFT JOIN Payment p
    ->     ON a.Student_ID = p.Student_ID
    -> GROUP BY h.Hostel_ID, h.Hostel_Name;
+-----------+-------------------+-------------------+
| Hostel_ID | Hostel_Name       | Total_Fee_Revenue |
+-----------+-------------------+-------------------+
|         1 | Sunrise Hostel    |          90000.00 |
|         2 | Blue Ridge Hostel |         144000.00 |
|         3 | Lake View Hostel  |          40000.00 |
+-----------+-------------------+-------------------+
3 rows in set (0.00 sec)
```

10.

```
PES2UG24CS337@root [hostel]>SELECT
    ->     f.Facility_ID,
    ->     f.Facility_Name,
    ->     COUNT(u.Student_ID) AS Usage_Count
    -> FROM Facility f
    -> LEFT JOIN Uses u
    ->     ON f.Facility_ID = u.Facility_ID
    -> GROUP BY f.Facility_ID, f.Facility_Name
    -> ORDER BY f.Facility_ID;
+-------------+---------------+-------------+
| Facility_ID | Facility_Name | Usage_Count |
+-------------+---------------+-------------+
|           1 | WiFi Lounge   |           4 |
|           2 | Laundry Room  |           1 |
|           3 | Gym           |           3 |
|           4 | Study Room    |           2 |
|           5 | Common Room   |           2 |
+-------------+---------------+-------------+
5 rows in set (0.00 sec)
```

11.

```
PES2UG24CS337@root [hostel]>SELECT
    ->     f.Facility_ID,
    ->     f.Facility_Name,
    ->     h.Hostel_Name
    -> FROM Hostel h
    -> RIGHT JOIN Facility f
    ->     ON h.Hostel_ID = f.Hostel_ID;
+-------------+---------------+------------------+
| Facility_ID | Facility_Name | Hostel_Name      |
+-------------+---------------+------------------+
|           1 | WiFi Lounge   | Sunrise Hostel   |
|           2 | Laundry Room  | Sunrise Hostel   |
|           3 | Gym           | Blue Ridge Hostel |
|           4 | Study Room    | Blue Ridge Hostel |
|           5 | Common Room   | Lake View Hostel |
+-------------+---------------+------------------+
5 rows in set (0.00 sec)
```

12.

```
PES2UG24CS337@root [hostel]>SELECT
    ->     s.First_Name,
    ->     s.Last_Name,
    ->     s.Email AS Email_Address,
    ->     a.Room_Number
    -> FROM Student s
    -> LEFT JOIN Allocation a
    ->     ON s.Student_ID = a.Student_ID
    ->    AND a.Check_out_Date IS NULL;
+------------+-----------+----------------------+-------------+
| First_Name | Last_Name | Email_Address        | Room_Number |
+------------+-----------+----------------------+-------------+
| Aditya     | Verma     | aditya.verma@pes.edu |         101 |
| Neha       | Kulkarni  | neha.kulkarni@pes.edu |         101 |
| Rohan      | Shetty    | rohan.shetty@pes.edu |        NULL |
| Sanya      | Malhotra  | sanya.malhotra@pes.edu |         101 |
| Vikram     | Nair      | vikram.nair@pes.edu  |         102 |
| Ishita     | Bansal    | ishita.bansal@pes.edu |         102 |
| Arjun      | Reddy     | arjun.reddy@pes.edu  |         101 |
| Meera      | Pillai    | meera.pillai@pes.edu |        NULL |
| Kabir      | Oberoi    | kabir.oberoi@pes.edu |        NULL |
+------------+-----------+----------------------+-------------+
9 rows in set (0.00 sec)
```